

Introduction to Python

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Definition

Easy to interpret and high-level object-oriented programming language with easy-to-read syntax.



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Why Python?

- Simplicity
- Easy interface
- Big toolkit available (modules/packages)



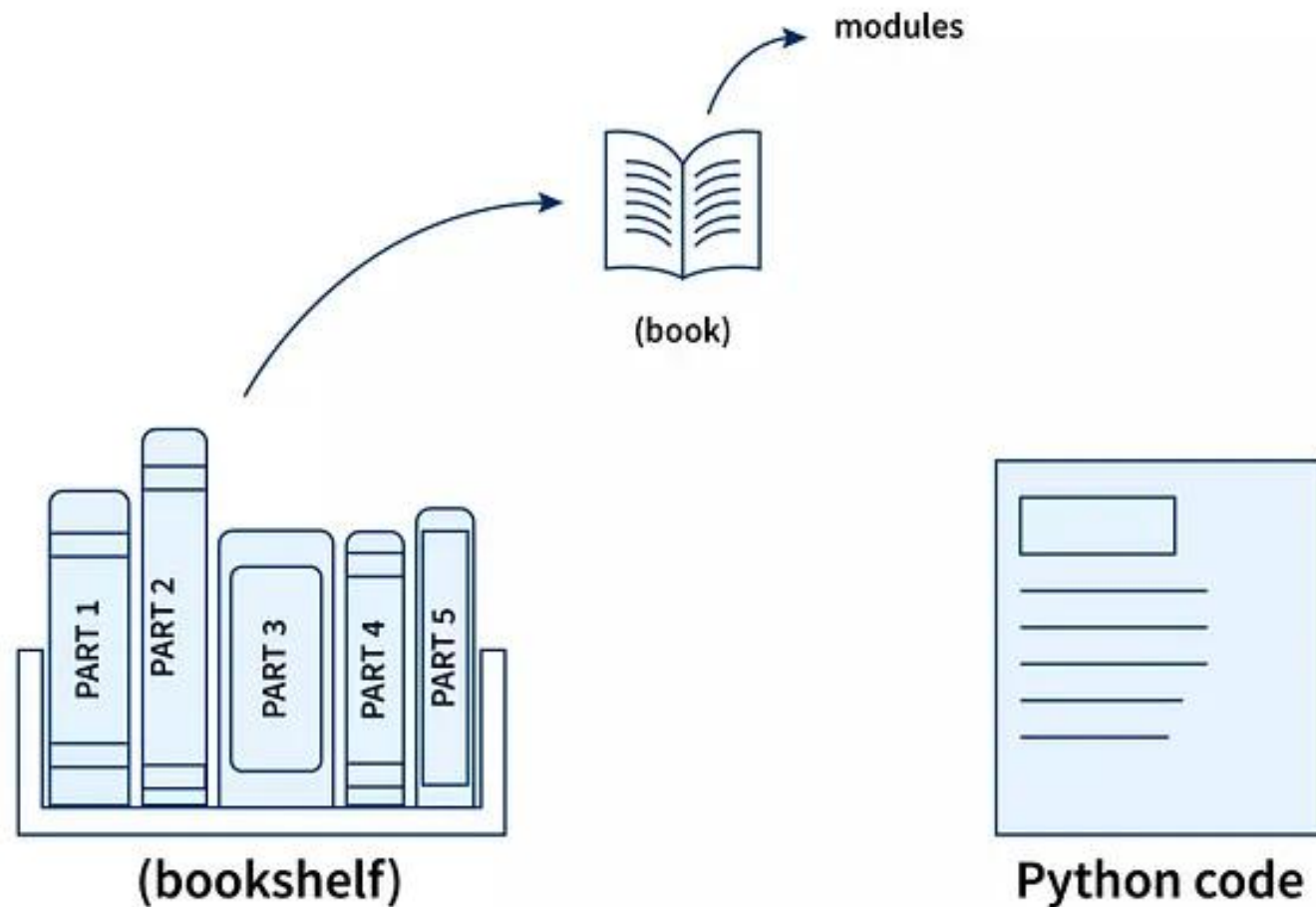
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Common scientific packages

Basics

- NumPy (array data)
- Pandas (tabular data)
- Matplotlib (data visualization)
- Seaborn (data visualization)

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Chemistry

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- Open Babel (conversion of chemical file formats)
- PyMOL (molecular graphics tool)
- ChemPy (physical chemistry)

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Machine Learning

- TensorFlow
- Scikit-learn

Virtual environments

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Mainly used:

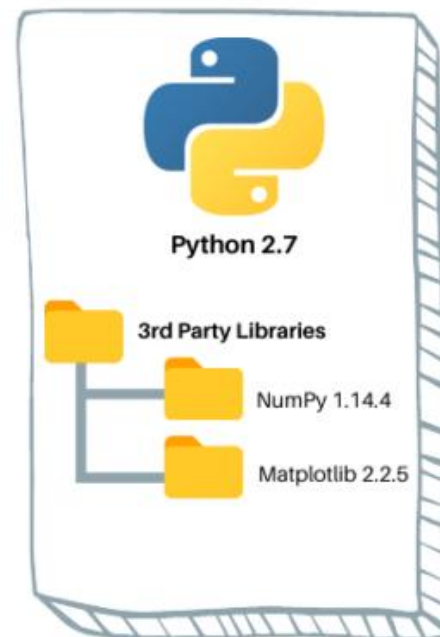
- to avoid dependency conflicts
- for reproducibility

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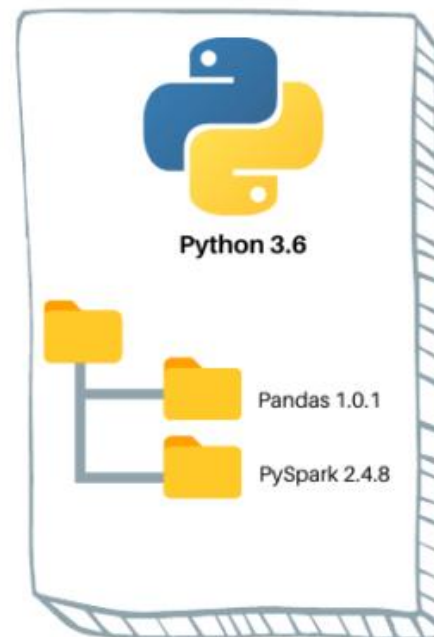
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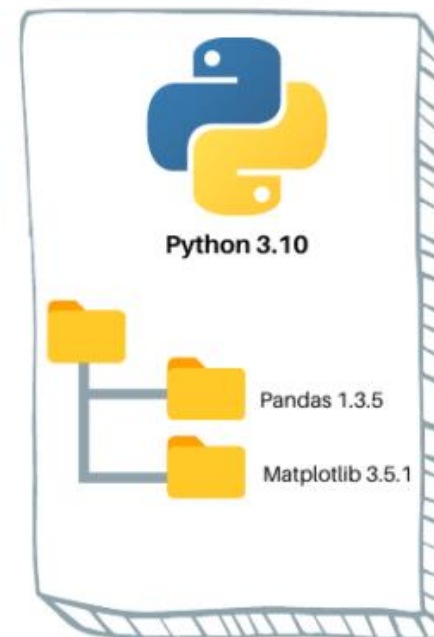
Virtual Environment 1



Virtual Environment 2



Virtual Environment 3



Virtual environment example

Virtual environment example

Python

Real life example

Virtual environment example

Python

Virtual environment

Real life example

Personal fumehood

Virtual environment example

Python

Virtual environment

Packages

Real life example

Personal fumehood

Reagents

Virtual environment example

Python

Virtual environment

Packages

Versions

Real life example

Personal fumehood

Reagents

Conductions

Virtual environment example

Python

Virtual environment

Packages

Versions

Requirements (yaml)

Real life example

Personal fumehood

Reagents

Conductions

Lab notebook

Virtual environment example

Python

Virtual environment

Packages

Versions

Requirements (yaml)

Real life example

Personal fumehood

Reagents

Conductions

Lab notebook

Other examples

- cheminf
- almos